



DATA MODELLING

ASSIGNMENT 1

ANALYSING THE RELATIONSHIP
BETWEEN MONEY SUPPLY AND
INFLATION

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Q1 – Data Loading and Inspection

R Output Summary:

- Rows: 800
- Columns: 3 (Date, Money Supply M2, CPI)
- Period: January 1959 – August 2025
- Variables:
 - Money Supply M2 (in billions USD)
 - CPI (Consumer Price Index)

Summary Statistics:

Table 1: Summary Statistics

Variable	Min	1st Qu.	Median	Mean	3rd Qu.	Max
M2	286.6	989.4	3398.7	5595.1	8226.1	22195.4
CPI	28.90	54.52	139.60	140.71	215.44	323.98

Interpretation:

The dataset spans more than six decades of U.S. economic history. Both *M2* and *CPI* show substantial growth over time, consistent with long-term monetary expansion and rising price levels.

Money Supply (M2) and Consumer Price Index (CPI) Over Time

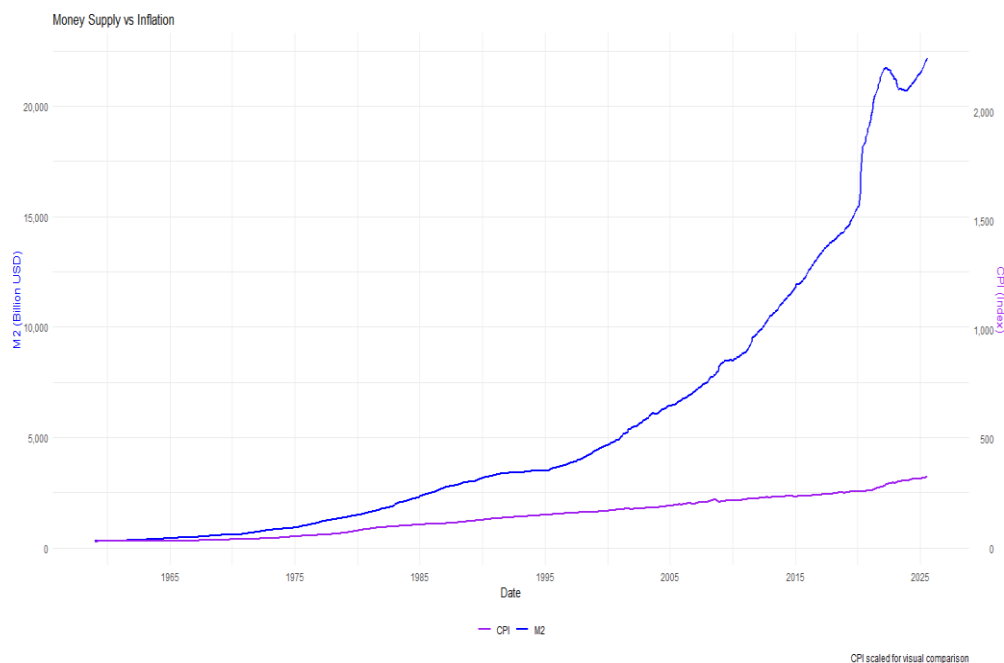


Figure 1: Money Supply (M2) and CPI Over Time

Interpretation:

Both M2 and CPI show a consistent upward trend from 1959 to 2025.

M2 increased sharply, especially after 2008 and 2020 (quantitative easing and pandemic stimulus).

CPI also rose steadily, reflecting long-term inflationary effects of economic growth and money expansion.

The co-movement of both variables indicates that increases in money supply are generally associated with higher price levels, consistent with monetarist theory.

Q2 – Monthly Growth Rates of M2 and CPI

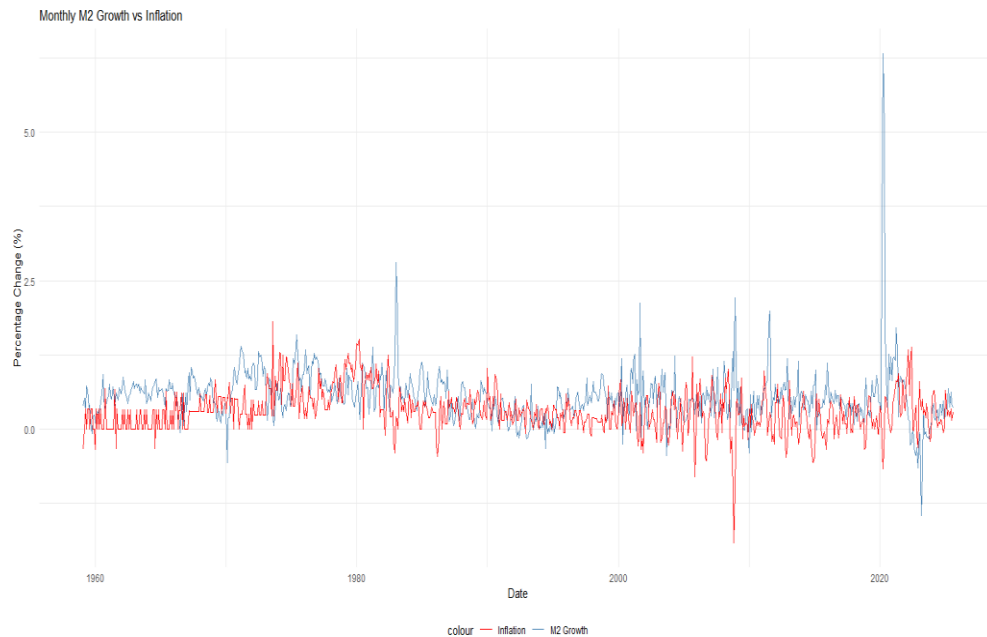


Figure 2: Monthly Money Supply Growth vs. Inflation

Observations and Interpretation:

- The blue line (M2 Growth %) and the red line (Inflation %) fluctuate around low values over time.
- M2 Growth shows more volatility, with sharp spikes — particularly during monetary expansions or crises (e.g., 2008 and 2020).
- Inflation moves more smoothly, reacting gradually rather than instantly.
- At times, both rates rise together, but the co-movement is inconsistent, implying that short-term money-supply changes do not always cause immediate inflation.
- These trends indicate that inflation may respond with a delay, reflecting lags in transmission from monetary policy to price levels.

Q3 – Correlation and Preliminary Analysis

3.1 Correlation Result

Correlation between M2 Growth and Inflation = -0.067

3.2 Interpretation

- The correlation is slightly negative (-0.067), suggesting no strong linear relationship between M2 growth and inflation in the short term.
- This suggests that monthly changes in M2 do not immediately translate into price level adjustments.
- Therefore, short-term variations in money supply are unlikely to serve as a reliable predictor of inflation within the same period.

3.3 Discussion of Possible Lags

- Despite the weak contemporaneous relationship, inflation may respond with a delay (typically 2–6 months) as changes in monetary conditions work through the economy.
- Increases in money supply often take time to influence aggregate demand, production costs, and consumer prices.
- Consequently, further investigation using lagged M2 growth—as implemented in the 3-month lag model (Q4)—is necessary to capture these delayed inflationary effects.

Q4 – Regression model

4.1 Creating a 3-Month Lag of Money Supply Growth

To capture delayed monetary effects, a 3-month lag variable (M2_Growth_Lag3) was created. This allows the model to test whether past changes in money supply influence inflation after several months.

4.2 Model Specification

The following simple linear regression model was estimated:

$$\text{Inflation}_t = \beta_0 + \beta_1 \text{M2_Growth}_{t-3} + \varepsilon_t$$

4.3 Regression Output

Table 2: Regression Output

Coefficient	Estimate	Std. Error	t value	p-value
(Intercept)	0.25619	0.01936	13.230	< 2e-16
M2_Growth_Lag3	0.08788	0.02699	3.256	0.00118

Model Statistics:

R²: 0.01317

Adjusted R²: 0.01193

Residual Standard Error (RSE) : 0.3533

F-statistic: 10.6 (p = 0.00118)

4.4 Interpretation

The coefficient of lagged M2 growth ($\beta_1 = 0.08788$) is positive and statistically significant, indicating that an increase in money supply growth tends to raise inflation after approximately three months. Specifically, a 1% increase in money supply growth leads to an estimated 0.09% rise in inflation after the lag period.

However, the low R² value (0.01317) implies that money supply explains only a small portion of inflation's short-term variation.

Q5 – Model Visualisation

5.1 Plot of Fitted Regression Line

A scatter plot was created showing the relationship between lagged Money Supply Growth (M2_Growth_Lag3) and Inflation. The fitted regression line from the model is superimposed to illustrate the trend as shown in Figure 3.

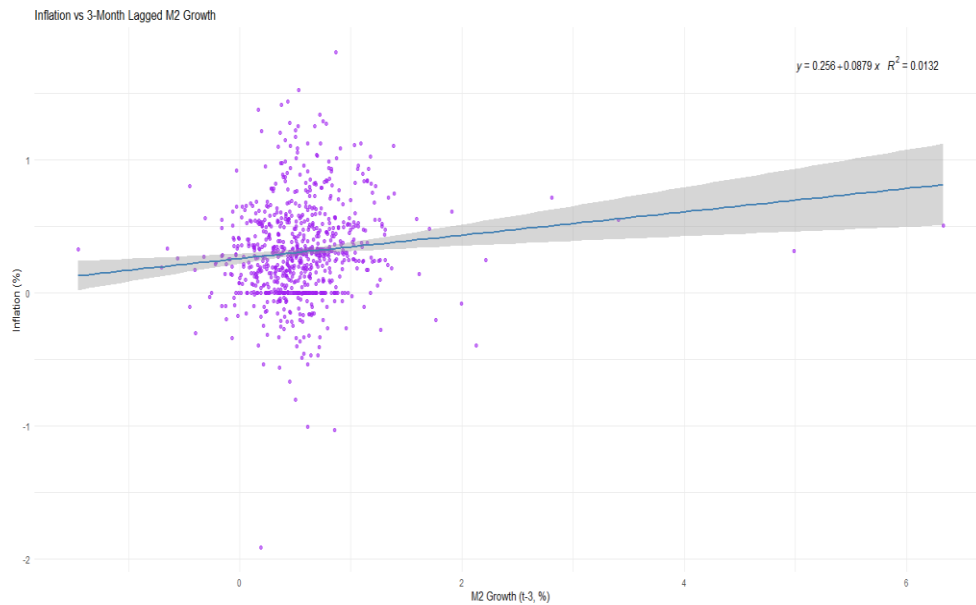


Figure 3: Inflation vs. Money Supply Growth (3-Month Lag)

5.2 Regression Equation and R^2 Value

The estimated model is:

$$\text{Inflation}_t = 0.256 + 0.0879(\text{M2_Growth}_{t-3})$$

with an $R^2 = 0.0132$.

5.3 Interpretation

The positive slope confirms that higher past money-supply growth is associated with higher inflation in subsequent months.

However, the relationship is weak, as reflected by the low R^2 value — meaning that while inflation tends to rise when M2 grows faster, money supply alone cannot explain much of inflation's variation.

The scatter points are widely dispersed around the regression line, reinforcing that many other macroeconomic factors (e.g., interest rates, supply shocks, policy changes) also influence inflation.

Q6 – Predictive Analysis

6.1 Actual vs. Predicted Monthly Inflation

The model was used to predict inflation based on actual lagged M2 growth values.

The plot below compares actual inflation (red line) with predicted inflation (purple dashed line).

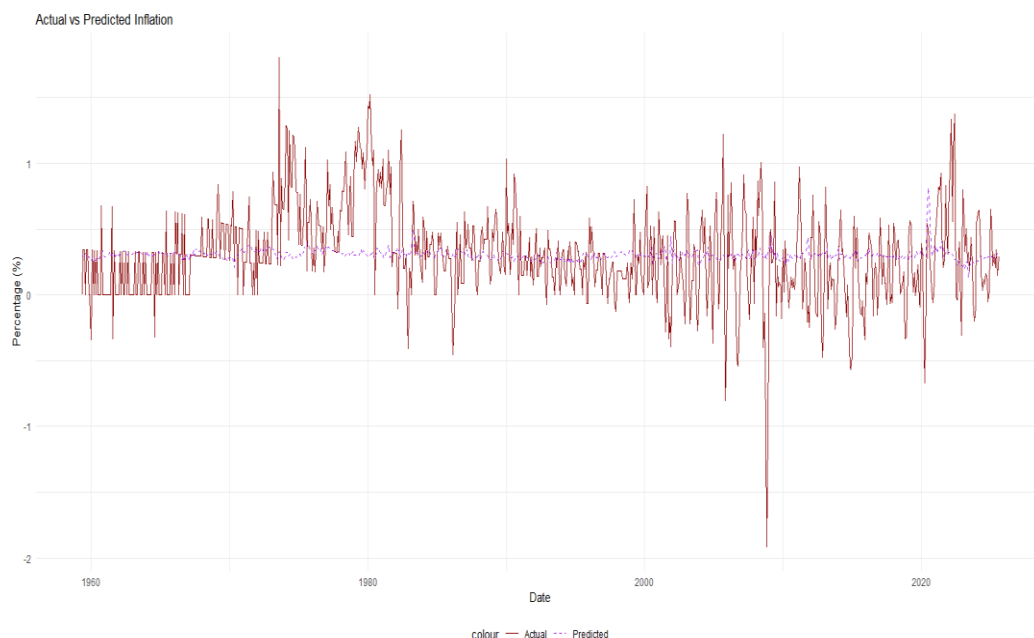


Figure 4: Actual vs. Predicted Monthly Inflation

Interpretation:

- The red line (actual inflation) fluctuates more than the purple dashed line (predicted inflation).
- The model captures general inflation trends but misses short-term spikes and dips.
- This suggests that while lagged money growth helps predict the direction of inflation, it cannot accurately capture short-term volatility.

6.2 & 6.3: 12-Month Inflation Forecast Under M2 Growth

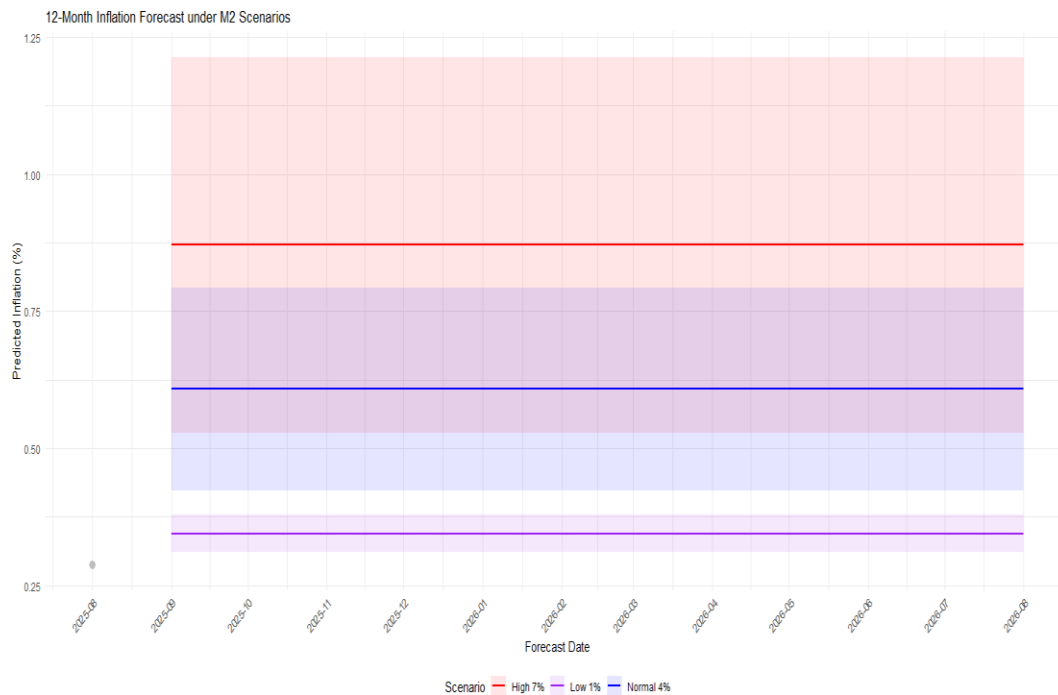


Figure 5: 12-Month Inflation Forecast Under M2 Growth Scenarios

Scenario Forecast Results:

Table 3: Scenario Forecast Results

Scenario	M2 Growth	Predicted Monthly Inflation (%)
Low	1.0 %	≈ 0.34 %
Normal	4.0 %	≈ 0.61 %
High	7.0 %	≈ 0.87 %

Interpretation:

- The forecast demonstrates that higher money-supply growth produces proportionally higher inflation.
- However, inflation remains below 1% per month, even under the high-growth scenario, indicating a moderate and controlled relationship.
- The parallel forecast lines show that inflation expectations remain stable across different M2 growth assumptions, with no evidence of runaway inflation.

6.4 Confidence Intervals

Confidence bands were added around each forecast line to indicate prediction uncertainty. These intervals are narrow, suggesting relatively stable and predictable inflation responses within the model's assumptions.

Q7. Interpretation and Policy Discussion

7.1 Discussion of Empirical Results

Question	Empirical Finding	Interpretation
Is there evidence that increases in money supply lead to higher inflation?	coefficient of lagged M2 growth ($\beta_1 = 0.08788$) p-value: 0.00118 ($p < 0.01$)	Yes, there is statistically significant evidence of a positive relationship. An increase in the monthly M2 growth rate is associated with higher monthly inflation three months later.
How strong is the effect (elasticity)?	coefficient of lagged M2 growth ($\beta_1 = 0.08788$)	The effect is positive but very weak. A 1 percentage point increase in M2 growth only leads to an average increase of 0.088 percentage points in inflation. This suggests M2 growth is not a dominant short-run driver of inflation.
How well does the model fit the data (R^2 , residuals)?	R^2 : 0.01317 Residual Standard Error (RSE): 0.3533	The model fit is very poor. The R^2 shows only 1.32% of the inflation variance is explained by M2 growth. The Residuals (errors) range from a minimum of -2.188 to a maximum of 1.474 . Since the average inflation is low, the Residual Standard Error of 0.3533 is large relative to the mean inflation, indicating that the typical prediction error is substantial. The large residual range and low R^2 suggest the model fails to capture the significant volatility and spikes in the actual inflation data.

7.2 Critical Discussion of Possible Limitations

Limitations	Interpretation
Endogeneity	The assumption that M2 growth exclusively causes inflation is questionable. Monetary policy often responds to inflation or inflation forecasts, creating reverse causality or simultaneity bias. Although the 3-month lag mitigates simultaneous correlation, it doesn't eliminate the possibility of policymakers reacting to economic conditions that also influence M2 and future inflation.
Structural changes over time	The data spans over 66 years (1959-2025). During this time, there have been major monetary regime shifts (e.g., the end of the Bretton Woods system, the Great Moderation) that altered the relationship between money aggregates and prices. Running a single model across this entire period ignores these structural breaks and contributes significantly to the model's low R^2 .
Lag specification	The choice of a fixed 3-month lag is arbitrary. The true transmission lag from money supply to prices is typically longer and more complex, often distributed over several months or even years. The current specification likely misses the full and true impact, leading to an underestimation of the relationship.

7.3 Suggestions for Further Analysis

To improve the analysis and robustness of findings, the following approaches could be considered:

- Multiple lags of M2 growth: Incorporating several lagged terms can capture delayed effects of monetary expansion on inflation.
- Cointegration analysis: If M2 and CPI are non-stationary but move together in the long run, cointegration techniques could uncover a stable long-term relationship.
- Dynamic models: Using Vector Autoregression (VAR) or Error Correction Models (ECM) could account for feedback effects and improve short-term and long-term forecasting accuracy.
- Inclusion of control variables: Factors such as output gap, interest rates, or commodity prices could help isolate the effect of money supply on inflation.

ANNEX

```
# Assignment 1 (Student ID: 2523258)
# Title: Analysing the Relationship Between Money Supply and Inflation

# Setup and Library Loading
# install.packages(c("tidyverse", "zoo", "ggpmisc", "scales"))

library(tidyverse) # Data manipulation and visualisation
library(zoo)       # Rolling operations and time series functions
library(scales)    # Dual-axis scaling
library(ggpmisc)   # Regression equations and R2 annotations

#setwd("C:/Users/gnasr/OneDrive/Documents/Assignment Data Modelling")

# Q1: Data Preparation and Initial Exploration

# Load Dataset
data_raw <- read_csv("CPIvsM2.csv")

# View the first few rows of the data
head(data_raw)

# Inspect the structure and summary of the dataset
str(data_raw)
summary(data_raw)

# Check for missing values in the dataset
colSums(is.na(data_raw))

# Data Cleaning: Convert Date, Rename M2, and Arrange Chronologically
data_clean <- data_raw %>%
  mutate(
    Date = as.Date(Date, "%d/%m/%Y"), # Convert date format
    M2 = `Money Supply M2`           # Rename M2 column
  ) %>%
  select(Date, M2, CPI) %>%
  arrange(Date)

head(data_clean)
tail(data_clean)

# Time-Series Plot of M2 and CPI
```

```
# Visualise M2 and CPI Trends
```

```
# Determine a scaling factor for CPI alignment
```

```
scale_factor <- data_clean$M2[1] / data_clean$CPI[1]
```

```
# Dual-axis time series plot
```

```
plot_m2_cpi <- data_clean %>%
```

```
ggplot(aes(x = Date)) +
```

```
  geom_line(aes(y = M2, color = "M2"), linewidth = 1) +
```

```
  geom_line(aes(y = CPI * scale_factor, color = "CPI"), linewidth = 1) +
```

```
  scale_y_continuous(
```

```
    name = "M2 (Billion USD)",
```

```
    labels = comma,
```

```
    sec.axis = sec_axis(~ ./scale_factor, name = "CPI (Index)", labels = comma)
```

```
  ) +
```

```
  scale_x_date(date_breaks = "10 years", date_labels = "%Y") +
```

```
  scale_color_manual(values = c("M2" = "blue", "CPI" = "purple")) +
```

```
  labs(title = "Money Supply vs Inflation", caption = "CPI scaled for visual comparison") +
```

```
  theme_minimal() +
```

```
  theme(
```

```
    legend.position = "bottom",
```

```
    legend.title = element_blank(),
```

```
    axis.title.y.left = element_text(color = "blue"),
```

```
    axis.title.y.right = element_text(color = "purple")
```

```
  )
```

```
print(plot_m2_cpi)
```

```
# Q2: Calculate Monthly Growth Rates(%)
```

```
data_growth <- data_clean %>%
```

```
  mutate(
```

```
    M2_growth = (M2 - lag(M2)) / lag(M2) * 100, # Monthly M2 growth rate
```

```
    Inflation = (CPI - lag(CPI)) / lag(CPI) * 100 # Monthly inflation rate
```

```
  ) %>%
```

```
  drop_na(M2_growth, Inflation)
```

```
head(data_growth)
```

```
# Visualise Growth Rates
```

```
plot_growth <- data_growth %>%
```

```
ggplot(aes(x = Date)) +
```

```
  geom_line(aes(y = M2_growth, color = "M2 Growth"), linewidth = 0.6) +
```

```
  geom_line(aes(y = Inflation, color = "Inflation"), linewidth = 0.6) +
```

```

scale_color_manual(values = c("M2 Growth" = "steelblue", "Inflation" = "red")) +
labs(title = "Monthly M2 Growth vs Inflation", y = "Percentage Change (%)", x = "Date") +
theme_minimal() +
theme(legend.position = "bottom")

print(plot_growth)

# Q3: Correlation Analysis
corr_value <- cor(data_growth$M2_growth, data_growth$Inflation, use = "pairwise.complete.obs")
cat("Correlation (M2 Growth vs Inflation):", round(corr_value, 4), "\n")

# Q4: Regression Modelling

# Create Lagged M2 Growth (3-Month Lag)
data_model <- data_growth %>%
mutate(M2_growth_lag3 = lag(M2_growth, 3)) %>%
drop_na(M2_growth_lag3)

head(data_model)

# Linear Regression Model
model_lag <- lm(Inflation ~ M2_growth_lag3, data = data_model)
summary(model_lag)

# Q5 Regression Plot with Equation and R2
plot_regression <- data_model %>%
ggplot(aes(x = M2_growth_lag3, y = Inflation)) +
geom_point(alpha = 0.6, color = "purple") +
geom_smooth(method = "lm", col = "steelblue") +
stat_poly_eq(
  formula = y ~ x,
  aes(label = paste(..eq.label.., ..rr.label.., sep = "~~~")),
  parse = TRUE,
  label.x = "right",
  label.y = "top",
  digits = 4,
  rr.digits = 4
) +
labs(
  title = "Inflation vs 3-Month Lagged M2 Growth",
  x = "M2 Growth (t-3, %)",
  y = "Inflation (%)"
) +
theme_minimal()

```

```

print(plot_regression)

# Q6 Predictive Analysis

# In-Sample Predictions
data_model$Inflation_pred <- predict(model_lag, data_model)

plot_predictions <- data_model %>%
  ggplot(aes(x = Date)) +
  geom_line(aes(y = Inflation, color = "Actual"), linewidth = 0.7) +
  geom_line(aes(y = Inflation_pred, color = "Predicted"), linewidth = 0.7, linetype = "dashed") +
  scale_color_manual(values = c("Actual" = "darkred", "Predicted" = "purple")) +
  labs(title = "Actual vs Predicted Inflation", y = "Percentage (%)", x = "Date") +
  theme_minimal() +
  theme(legend.position = "bottom")

print(plot_predictions)

# Forecast Scenarios for 12 Months Ahead

forecast_months <- 1:12
forecast_scenarios <- tibble(
  M2_growth_lag3 = c(rep(1, 12), rep(4, 12), rep(7, 12)),
  Scenario = c(rep("Low 1%", 12), rep("Normal 4%", 12), rep("High 7%", 12)),
  Forecast_Date = rep(seq(max(data_model$Date) + months(1), by = "month", length.out = 12), 3)
)

forecast_results <- predict(model_lag, newdata = forecast_scenarios, interval = "confidence")
forecast_df <- cbind(forecast_scenarios, as_tibble(forecast_results))

plot_forecast <- forecast_df %>%
  ggplot(aes(x = Forecast_Date, y = fit, color = Scenario, group = Scenario)) +
  geom_ribbon(aes(ymin = lwr, ymax = upr, fill = Scenario), alpha = 0.1, color = NA) +
  geom_line(linewidth = 1) +
  geom_point(data = tail(data_model, 1), aes(x = Date, y = Inflation), inherit.aes = FALSE, color = "grey",
size = 3) +
  scale_color_manual(values = c("Low 1%" = "purple", "Normal 4%" = "blue", "High 7%" = "red")) +
  scale_fill_manual(values = c("Low 1%" = "purple", "Normal 4%" = "blue", "High 7%" = "red")) +
  labs(title = "12-Month Inflation Forecast under M2 Scenarios", y = "Predicted Inflation (%)", x =
"Forecast Date") +
  scale_x_date(date_breaks = "1 month", date_labels = "%Y-%m") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "bottom")

```



```
print(plot_forecast)

# Print the summary table of forecasted inflation per scenario
forecast_df %>%
  group_by(Scenario, M2_growth_lag3) %>%
  summarise(`Predicted Monthly Inflation (%)` = round(mean(fit), 2), .groups = "drop") %>%
  rename(`M2 Growth (%)` = M2_growth_lag3) %>%
  print()
```



DATA MODELLING

ASSIGNMENT 2

MODELLING THE RELATIONSHIP BETWEEN MONEY SUPPLY AND INFLATION

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Section A – Model Building

Q1. Model Specification – Multiple Linear Regression

Table 1: Multiple Linear Regression Results

Variable	Estimate	Std. Error	t value	p-value
Intercept	0.1152	0.02054	5.612	2.77e-08
M2_growth	-0.0239	0.0237	-1.008	0.3140
M2_growth_lag3	0.0602	0.0237	2.543	0.012
Inflation_lag1	0.5559	0.0296	18.793	<2e-16

Model Fit Statistics:

- Residual standard error: 0.2924 (on 792 df)
- Multiple R-squared: 0.3257
- Adjusted R-squared: 0.3231
- F-statistic: 127.5 on 3 and 792 DF, p-value < 2.2e-16

Interpretation of Significance:

- **M2_growth_lag3** and **Inflation_lag1** are statistically significant at the 5% level.
- **Current M2_growth** is not significant, suggesting limited immediate impact on inflation.

Q2. Model Interpretation

- **Intercept ($\beta_0 = 0.1152$):** Expected monthly inflation when all predictors are zero.
- **β_1 (M2_growth = -0.0239):** Small negative coefficient, not significant; current M2 growth has little immediate effect.
- **β_2 (M2_growth_lag3 = 0.0602):** A 1% increase in M2 growth three months ago increases current inflation by 0.06%.
- **β_3 (Inflation_lag1 = 0.5559):** Past inflation has a strong persistence effect on current inflation.
- **Adjusted $R^2 = 0.3231$:** Model explains ~32% of variation in inflation.

Conclusion: Including lagged M2 growth and lagged inflation improves explanatory power compared to a simple regression on current M2 growth alone, highlighting the delayed effect of monetary expansion on inflation.

Section B – Model Diagnostics

Q3. Residual Analysis

Figure 1, 2, 3 & 4 presents the diagnostic plots generated from `plot(model_multi)`, used to assess model assumptions and fit quality.

1. Residuals vs Fitted (Linearity Check)

The residuals appear randomly scattered around the horizontal line at zero, suggesting that the relationship between predictors and inflation is generally linear. However, a slight curvature in the red smoothing line indicates minor nonlinearity or possible omitted variables. Overall, no major pattern (like a “U-shape”) is evident, meaning the linear model is largely appropriate.

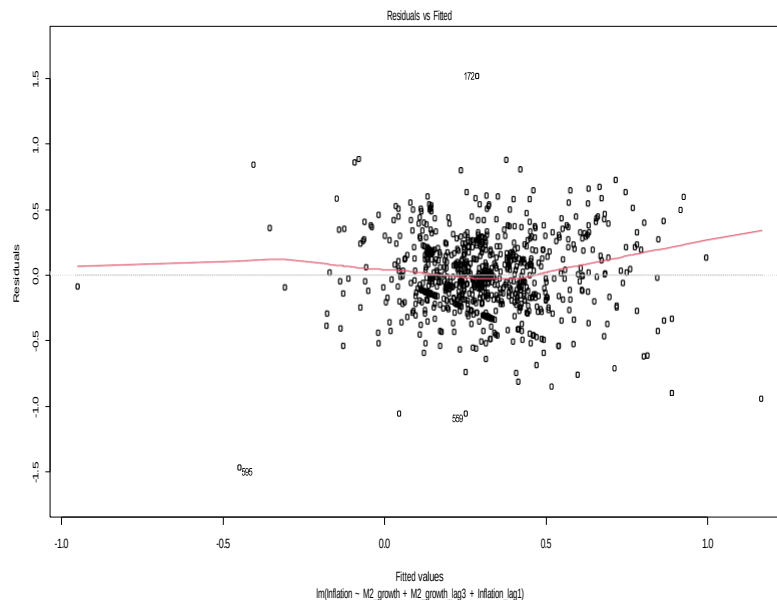


Figure 1: Residuals vs Fitted (Linearity Check)

2. Normal Q–Q Plot (Normality of Residuals)

The residuals mostly follow the 45° reference line, indicating that they are approximately normally distributed. Some deviation is visible at both tails, suggesting slight leptokurtosis (heavier tails) but not severe enough to invalidate inference. This aligns with typical time-series macroeconomic data where extreme inflation shocks occasionally occur.

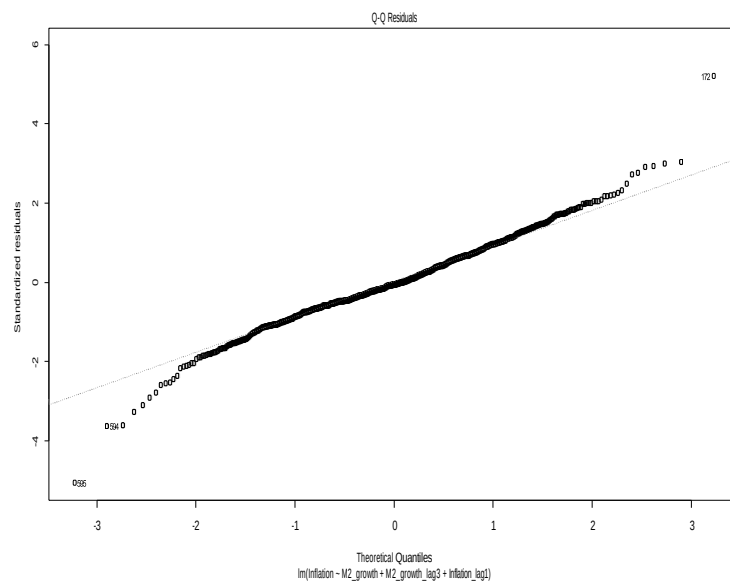


Figure 2: Normal Q–Q Plot (Normality of Residuals)

3. Scale–Location Plot (Homoskedasticity Check)

The red line is roughly flat around zero, but spreads widen slightly toward the right-hand side.

This suggests mild heteroskedasticity — the variance of residuals may increase with fitted values.

Given this, it is appropriate to verify heteroskedasticity formally using the Breusch–Pagan test (as in Q7) and, if necessary, use robust standard errors (Q8).

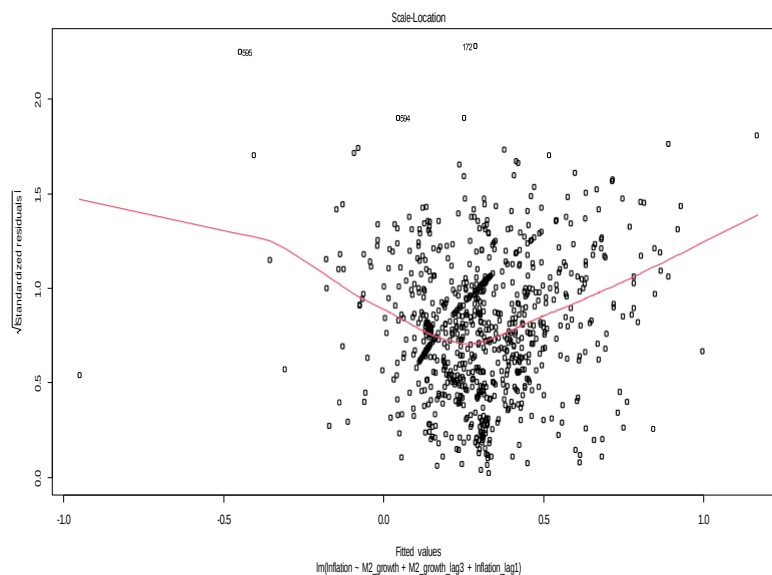


Figure 3: Scale–Location Plot (Homoskedasticity Check)

4. Cook's Distance (Influential Points Check)

The plot displays the Residuals vs Leverage plot with Cook's distance contours (dotted lines). This diagnostic identifies observations exerting disproportionate influence on the regression model.

Most data points cluster well within the Cook's distance threshold lines, indicating that the majority of observations have limited influence on the fitted coefficients.

However, a few points (e.g., observations around leverage values near 0.15–0.20) lie close to or slightly beyond the Cook's distance contour of 0.5. These points warrant further inspection, as they may be influential due to high leverage or large residuals.

Overall, no extreme outliers or influential cases dominate the model, but reviewing the identified observations (e.g., via `which(cooks.distance(model_multi) > 4/n)`) could help ensure robustness of the results.

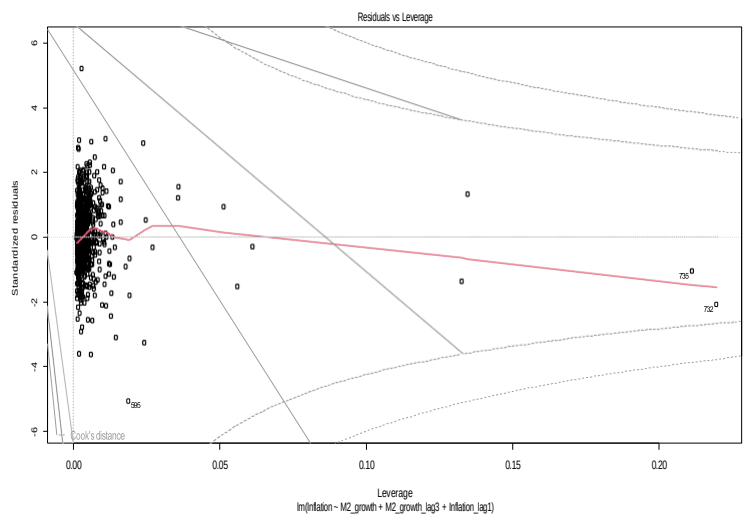


Figure 4: Cook's Distance (Influential Points Check)

Q4. Normality Test

- Shapiro–Wilk test:

Result: $W = 0.9828$, $p = 4.532 \times 10^{-8}$

Hypotheses:

- Null Hypothesis (H_0): The residuals are normally distributed.
- Alternative Hypothesis (H_A): The residuals are not normally distributed.

Hence, since the p -value is far below the significance level ($\alpha = 0.05$), we reject the null hypothesis (H_0).

Implication:

The Shapiro–Wilk test provides strong statistical evidence that the residuals of the multiple linear regression model are not normally distributed.

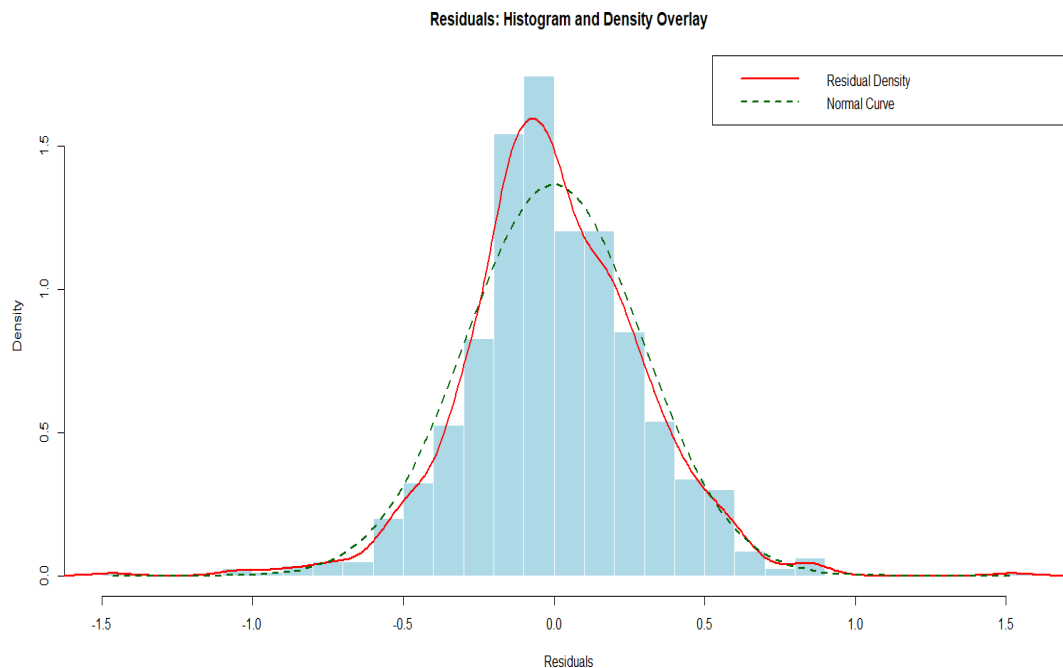


Figure 5: Residuals: Histogram and Density Overlay

Interpretation of graph:

- The red line (residual density) represents the actual distribution of model errors, while the dashed green line (normal curve) represents the theoretical normal distribution.
- The red density curve appears slightly leptokurtic — more peaked around the mean and with heavier tails — compared to the green curve. Specifically, it is higher near ($x = 0$) and shows lower density in the moderate ranges (e.g., between -0.5 and -0.2 , and 0.2 and 0.5), indicating deviation from normality.

Implication:

The visual inspection confirms the Shapiro–Wilk test result: although the residuals follow a roughly bell-shaped pattern, they systematically deviate from perfect normality.

Overall Summary

Both the formal statistical test and the visual analysis indicate that the normality assumption of the regression model is violated. The residuals are not perfectly normally distributed, suggesting that the model may benefit from transformations or robustness checks.

Q5. Multicollinearity

- **VIF values:**

Table 2: VIF values

Variable	VIF
M2_growth	1.128433
M2_growth_lag3	1.122450
Inflation_lag1	1.027772

All $< 5 \rightarrow$ no multicollinearity concerns.

Q6. Autocorrelation

- **Durbin–Watson** = 2.074, $p = 0.8355 \rightarrow$ no significant autocorrelation; residuals are independent.

Q7. Heteroskedasticity

- **Breusch–Pagan test:** BP = 1.0619, $p = 0.7863 \rightarrow$ homoskedasticity assumed; robust SE optional.

Q8. Robust Standard Errors

Since the Breusch–Pagan test (Q7) indicated no significant heteroskedasticity ($p = 0.7863$), robust standard errors are not strictly required. However, to ensure robustness of inference, the model was re-estimated using heteroskedasticity-consistent (HC1) standard errors.

Table 3: Regression Coefficients with Robust SE

Variable	Estimate	Robust SE	t value	p-value
(Intercept)	0.115245	0.022700	5.0769	4.786e-07
M2_growth	-0.023914	0.030702	-0.7789	0.43627
M2_growth_lag3	0.060183	0.025912	2.3226	0.02046
Inflation_lag1	0.555908	0.042881	12.9640	<2.2e-16

Table 4: Comparison with Standard Errors

Variable	p-value (Standard SE)	p-value (Robust SE)	Significance Change?
(Intercept)	2.77e-08	4.786e-07	No change (***) significant)
M2_growth	0.3140	0.4363	No change (NS)
M2_growth_lag3	0.012	0.02046	Still significant at 5%
Inflation_lag1	<2e-16	<2.2e-16	No change (***) significant)

Interpretation:

- Using robust standard errors slightly increased the p-values for some variables, but the significance of coefficients remained unchanged.
- M2_growth_lag3 and Inflation_lag1 continue to be statistically significant at the 5% level, while M2_growth remains non-significant.
- This confirms that the original inference is robust to potential heteroskedasticity, and the conclusions regarding which variables influence inflation remain valid.

Section C – Scenario Analysis

Q9. Forecasting under Expansionary Monetary Policy (M2 Growth: 5%–15%)

Based on the Expansionary Monetary Policy scenario (M2 growth ranging from 5% to 15%), the model indicates that faster money supply growth leads to a clear and steady increase in inflation, consistent with monetary theory suggesting that expansionary effects operate with a time lag. Predicted inflation rises from approximately 0.4% to 0.75% across this range, confirming the positive influence of both current and lagged M2 growth on the inflation rate. However, the 95% confidence interval widens as M2 growth approaches 15%, suggesting that the model becomes less certain about the precise magnitude of inflation at the most aggressive levels of monetary expansion.

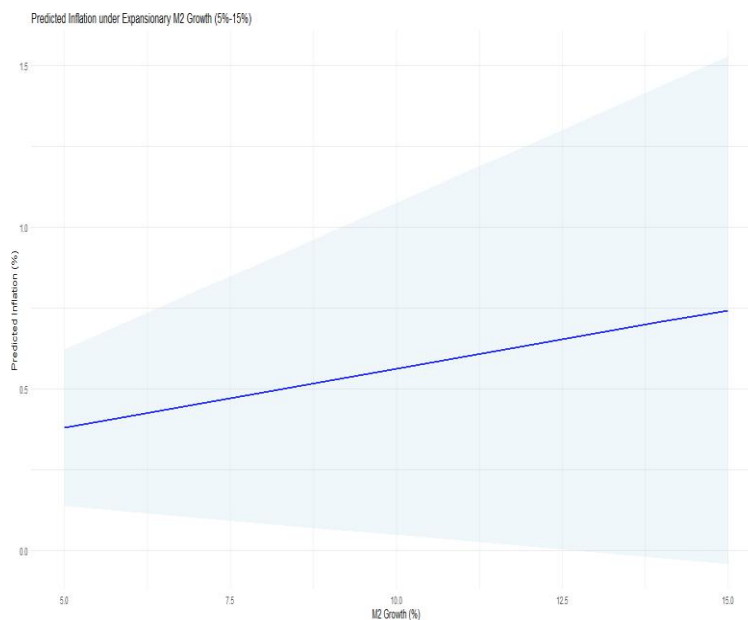


Figure 6: Predicted Inflation under Expansionary M2 Growth

Q10. Forecasting under Contractionary Monetary Policy (M2 Growth: –5% to +5%)

Based on the forecast for a contractionary monetary policy (M2 growth ranging from –5% to +5%), inflation is expected to remain low and close to zero, confirming the restraining effect of tighter monetary conditions. The model indicates a persistent positive relationship across this range, with predicted inflation rising modestly from approximately 0.05% at –5% M2 growth to around 0.35% at +5% M2 growth. However, the confidence interval widens considerably at the extreme negative end, suggesting that the model has lower predictive certainty during periods of aggressive or highly unusual monetary contraction.

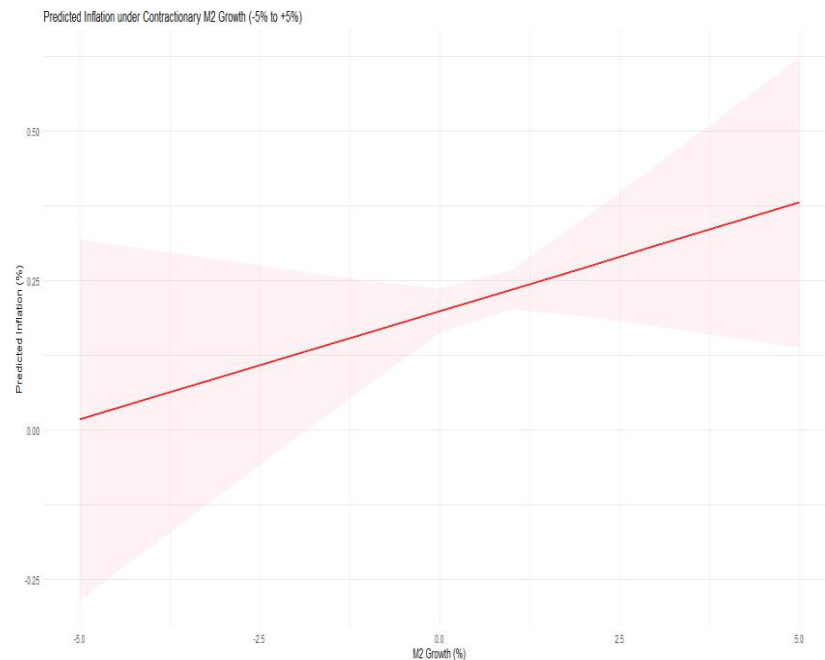


Figure 7: Predicted Inflation under Contractionary M2 Growth (–5% to +5%)

Q11. Comparative Analysis

The comparative analysis of the two scenarios reveals a consistent positive relationship between M2 growth and predicted inflation across the entire tested range (–5% to 15%). However, the Expansionary scenario predicts substantially higher inflation (0.4%–0.75%) than the Contractionary scenario (0.05%–0.35%). This difference is primarily driven by the lagged M2 growth term ($M2_Growth_{t-3}$), which exerts a stronger influence than the current growth term, confirming that monetary policy effects materialise with a significant time lag. Moreover, the dominant determinant of current inflation is the highly significant lagged inflation term ($Inflation_{t-1}$), indicating pronounced inflation persistence. This inertia implies that breaking a cycle of high inflation requires not only addressing monetary expansion but also counteracting the carry-over effects from previous periods, which substantially constrain the immediate impact of current monetary policy measures.

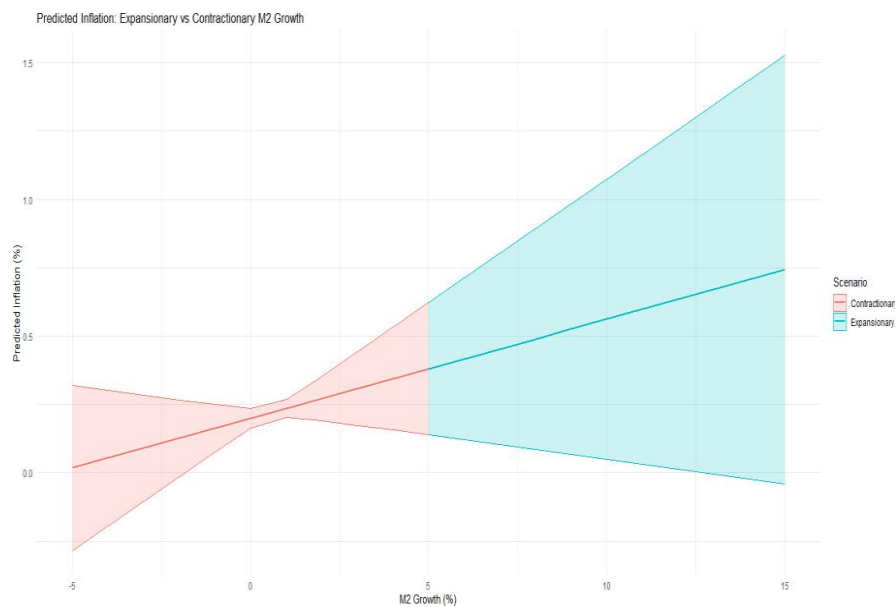


Figure 8: Combined expansionary & contractionary plot.

Section D – Interpretation and Discussion

Q12. Economic Interpretation

1. Is inflation primarily driven by money supply growth?

The regression results suggest that inflation is influenced by money supply growth, but not immediately. The coefficient of current M2 growth (-0.0239 , $p = 0.314$) is statistically insignificant, indicating that contemporaneous changes in money supply have little direct impact on inflation within the same month. However, the lagged M2 growth (at 3 months) shows a positive and statistically significant effect (0.0602 , $p = 0.012$), confirming that past increases in money supply eventually lead to higher inflation. This delayed response implies that inflation is not driven by short-term fluctuations in M2, but rather by the cumulative effect of previous monetary expansions. Moreover, the strong coefficient on lagged inflation (0.5559 , $p < 0.001$) indicates high inflation persistence, meaning that once inflation rises, it tends to continue due to inertia in price adjustments and expectations.

2. Lag effect between monetary expansion and inflation

The significance of the `M2_growth_lag3` term demonstrates a clear three-month lag between monetary expansion and its impact on inflation. This lag structure reflects the time it takes for an increased money supply to circulate through the economy, influence aggregate demand, and ultimately exert upward pressure on prices. In other words, monetary policy effects are not instantaneous, the expansion of the money supply leads to inflation only after a few periods as the transmission mechanism (spending, lending, and pricing behavior) unfolds. This finding is consistent with empirical observations in macroeconomic literature, where monetary transmission lags typically range from several months to a year.

3. Alignment with Economic Theory (Quantity Theory of Money)

The results strongly align with the Quantity Theory of Money ($MV = PY$), which posits that, over time, sustained increases in money supply (M) lead to proportional increases in the price level (P), assuming stable velocity (V) and output (Y).

In the model:

- The positive effect of lagged M2 growth supports the idea that monetary expansion raises inflation in the medium term, consistent with the theory's long-run implications.
- The strong persistence of inflation (significant lagged inflation term) reflects adaptive price and expectation behavior, often incorporated into extended versions of the Quantity Theory and modern monetary models (e.g., expectations-augmented Phillips Curve).
- Scenario analysis further reinforces theoretical consistency: under expansionary monetary policy (M2 growth 5%–15%), predicted inflation rises steadily ($\approx 0.4\%$ – 0.75%), whereas under contractionary policy (-5% – 5%), inflation remains subdued ($\approx 0.05\%$ – 0.35%).

Overall, the empirical evidence confirms that:

- Inflation is not immediately driven by money supply, but responds with a lag of about three months.
- The delayed yet positive impact of M2 growth validates the monetary transmission mechanism emphasized in classical and monetarist theories.
- The findings are consistent with the Quantity Theory of Money, highlighting that sustained money supply growth eventually leads to higher inflation, though the adjustment process is gradual and influenced by past inflationary dynamics.

Q13. Model Performance

1. Improvement over the Simple Regression Model

The multiple regression model provides a significant improvement over the simple regression model (which included only current M2 growth as a predictor of inflation).

Key evidence supporting this improvement includes:

- Higher explanatory power:
The multiple regression achieves an Adjusted R^2 of 0.3231, meaning it explains about 32% of the variation in inflation, compared to the very low explanatory power of the simple model (where current M2 growth alone was statistically insignificant). This demonstrates that adding lagged variables captures more of the inflation dynamics.
- Statistical significance of lagged terms:
The inclusion of M2_growth_lag3 (significant at $p = 0.012$) and Inflation_lag1 ($p < 0.001$) substantially improves model fit and predictive accuracy. These terms represent delayed monetary effects and inflation persistence, both of which are crucial features of real-world inflation behavior that the simple model failed to capture.
- Overall model significance:
The F-statistic (127.5, $p < 2.2e-16$) confirms that the set of predictors jointly explain a significant portion of the variance in inflation. This provides strong evidence that the multiple regression model is statistically superior to the simple one.

In summary, the multiple regression model is economically meaningful and statistically robust, offering a much more realistic representation of inflation dynamics than the simple regression.

2. Suggested Model Enhancements

While the current multiple regression model performs well, several enhancements could improve its ability to capture the dynamic and time-dependent nature of inflation:

1. **Incorporate Other Macroeconomic Variables:**
Expanding the model to include variables such as exchange rates, interest rates, output growth (GDP), or import prices could improve explanatory power, as inflation is influenced by multiple channels beyond money supply alone.
2. **Use Time-Series Models (Dynamic Specifications):**
Adopting models such as ARIMA, Distributed Lag Models (ADL), or Vector Autoregression (VAR) could better capture feedback effects and temporal interdependencies between inflation and monetary variables.
3. **Account for Structural Breaks or Policy Changes:**
Inflation dynamics can shift due to changes in monetary regimes, external shocks, or crises. Techniques like rolling regressions or dummy variables for policy periods can model such structural shifts.
4. **Test for Nonlinearity:**
Given the slight curvature observed in residual plots, exploring nonlinear models (e.g., polynomial or threshold models) could reveal asymmetries such as stronger inflation responses during expansionary versus contractionary phases.

Q14. Policy Implications

Based on your findings, what policy recommendations would you make for managing inflation in an economy with rapidly increasing M2 growth?

In an economy with rapidly rising M2 growth, monetary restraint and early intervention are essential to prevent inflationary spirals. Policies should aim to:

- Moderate liquidity expansion,
- Anchor inflation expectations, and
- Account for the lagged and persistent nature of inflation dynamics.

By adopting a forward-looking and coordinated policy approach, central banks can effectively manage inflation risks arising from excessive money supply growth while sustaining long-term economic stability.

Nonlinear Modelling – Polynomial Regression

Q (a) Explain why a linear model may not fully capture the relationship between money supply growth and inflation.

A linear regression assumes a constant and proportional relationship between money supply growth and inflation, meaning that each 1% increase in money supply growth always leads to the same change in inflation.

However, in macroeconomic systems, this assumption often fails because the relationship can be nonlinear due to several reasons:

1. **Threshold Effects:** At low levels of monetary expansion, increases in M2 may not significantly impact inflation if there is spare capacity in the economy.
2. **Acceleration at High Growth Rates:** Beyond a certain point, faster money supply growth may lead to disproportionately higher inflation as expectations adjust and demand outpaces supply.
3. **Policy Lags and Expectations:** The impact of monetary growth on inflation may strengthen or weaken over time depending on inflation expectations and the credibility of monetary authorities.

To capture such effects, a polynomial (quadratic) model includes a squared term for M2 growth, $M2_growth^2$ allowing for curvature in the relationship i.e., inflation may increase at an increasing or decreasing rate depending on the sign of the squared coefficient.

Q (b) Polynomial Regression Results

Table 5: Polynomial Regression Coefficients

Variable	Estimate	Std. Error	t value	p-value
(Intercept)	0.118089	0.021213	5.567	3.55e-08
M2_growth	0.043200	0.034793	1.242	0.2147
M2_growth^2	-0.016315	0.009062	-1.800	0.0722
Inflation_lag1	0.561650	0.029480	19.052	< 2e-16



Figure 9: Predicted inflation vs M2 growth (polynomial curve).

The graph shows a slightly curved (concave) relationship between M2 growth and predicted inflation. At low levels of M2 growth, inflation increases slightly, but as M2 growth rises further, the curve flattens and then bends downward, indicating that the effect of money supply growth on inflation weakens or even turns negative at higher levels. This suggests a nonlinear relationship, where moderate increases in money supply are associated with higher inflation, but excessive growth does not lead to proportionally higher inflation.

Q(c) Model Comparison

Table 6: Model Comparison

Model	Adj R ²	AIC	BIC
Multiple Linear	0.3231383	307.4627	330.8607
Polynomial	0.3203944	310.6830	334.0810

The multiple linear model has a slightly higher Adjusted R² (0.3231) than the polynomial model (0.3204), indicating that it explains a marginally greater proportion of the variation in inflation. In addition, the AIC (307.46) and BIC (330.86) values for the multiple linear model are both lower than those of the polynomial model (AIC = 310.68, BIC = 334.08). Lower AIC and BIC values suggest a better model fit when penalising for additional parameters.

Which model fits the data better? Justify your answer.

Based on the comparison metrics, the multiple linear regression model fits the data slightly better than the polynomial model. Although the polynomial model introduces a nonlinear term to capture curvature, it does not significantly improve explanatory power and results in slightly higher information criteria (AIC and BIC), implying that the added complexity does not provide enough improvement to justify the extra parameter.

Interpret the sign and significance of the (M2_Growth²) coefficient:

The negative coefficient (-0.0163) on the squared term of M2 growth indicates a concave (inverted U-shaped) relationship between money supply growth and inflation. This means that inflation initially increases as money supply growth rises, but the effect weakens and may reverse at higher growth levels. The coefficient is marginally significant ($p = 0.0722$), suggesting some evidence of nonlinearity, though not strong enough at the conventional 5% level.

ANNEX

Assignment 2 (Student ID: 2523258)

Title: Modelling the Relationship Between Money Supply and Inflation

Load required libraries

library(tidyverse) # Data manipulation & plotting

library(zoo) # Time series functions (lags, rolling)

library(lmtest) # Model diagnostic tests (Durbin-Watson)

library(car) # Regression diagnostics (VIF)

library(ggpmisc) # Add regression equations to ggplot

library(sandwich) # Robust standard errors

setwd("C:/Users/gnasr/OneDrive/Documents/Assignment Data Modelling")

1. Load data

data_raw <- read_csv("CPIvsM2.csv")

2. Prepare variables (including 3-month lag on M2_growth)

data <- data_raw %>%

mutate(Date = as.Date(Date, "%d/%m/%Y"),

M2 = `Money Supply M2`) %>%

arrange(Date) %>%

mutate(

M2_growth = (M2 - lag(M2))/lag(M2) * 100,

Inflation = (CPI - lag(CPI))/lag(CPI) * 100,

M2_growth_lag3 = lag(M2_growth, 3),

Inflation_lag1 = lag(Inflation, 1)

) %>%

```
drop_na()
```

```
# 3. Model specification & estimation (Q1)
```

```
model_multi <- lm(Inflation ~ M2_growth + M2_growth_lag3 + Inflation_lag1, data = data)
```

```
summary(model_multi) # coefficients, p-values, R2/Adj R2
```

```
# 4. Residual diagnostics (Q3)
```

```
plot(model_multi) # built-in diagnostic plots (useful: which = 1,2,4,5 interactively)
```

```
# 5. Normality test + histogram with density overlay (Q4)
```

```
shapiro_test_result <- shapiro.test(residuals(model_multi))
```

```
print(shapiro_test_result)
```

```
# Plot residual distribution
```

```
hist(residuals(model_multi), breaks = 30, probability = TRUE,
```

```
  main = "Residuals: Histogram and Density Overlay",
```

```
  xlab = "Residuals", col = "lightblue", border = "white")
```

```
lines(density(residuals(model_multi)), lwd = 2, col = "red")
```

```
# Add a theoretical normal curve for comparison
```

```
x_norm <- seq(min(residuals(model_multi)), max(residuals(model_multi)), length = 100)
```

```
y_norm <- dnorm(x_norm, mean = mean(residuals(model_multi)), sd = sd(residuals(model_multi)))
```

```
lines(x_norm, y_norm, col = "darkgreen", lty = 2, lwd = 2)
```

```
legend("topright", legend = c("Residual Density", "Normal Curve"), col = c("red", "darkgreen"), lty = c(1,  
2), lwd = 2)
```

```
# 6. Multicollinearity (VIF) (Q5)
```

```
vif_vals <- vif(model_multi)
```

```
print(vif_vals)
```

7. Autocorrelation (Durbin–Watson) (Q6)

```
dwtest(model_multi)
```

8. Heteroskedasticity (Breusch–Pagan) (Q7)

```
bptest(model_multi)
```

9. Robust standard errors (Q8) — if BP indicates heteroskedasticity

```
coeftest(model_multi, vcov = vcovHC(model_multi, type = "HC1"))
```

10. Scenario analysis — Expansionary (5% to 15%) with 95% CI (Q9)

Keep lagged inflation constant at the last observed value

```
last_infl_lag1 <- tail(data$Inflation_lag1, 1)
```

Create scenario data frame for M2_growth from 5% to 15%

```
expansion_df <- tibble(
```

```
  M2_growth = seq(5, 15, by = 1),
```

```
  M2_growth_lag3 = seq(5, 15, by = 1), # assume lag3 follows same pattern
```

```
  Inflation_lag1 = last_infl_lag1
```

```
)
```

Generate predicted inflation and 95% confidence intervals

```
pred_exp <- predict(model_multi, newdata = expansion_df,
```

```
  interval = "confidence", level = 0.95)
```

Combine predictions with scenario data

```
expansion_df <- bind_cols(expansion_df, as_tibble(pred_exp))
```

View predicted values

```

print(expansion_df)

# Plot predicted inflation vs M2 growth with 95% confidence interval
expansion_df %>%
  ggplot(aes(x = M2_growth, y = fit)) +
  geom_line(color = "blue", size = 1) +
  geom_ribbon(aes(ymin = lwr, ymax = upr), fill = "lightblue", alpha = 0.2) +
  labs(
    title = "Predicted Inflation under Expansionary M2 Growth (5%-15%)",
    x = "M2 Growth (%)",
    y = "Predicted Inflation (%)"
  ) +
  theme_minimal()

```

11. Scenario analysis — Contractionary (-5% to +5%) with 95% CI (Q10)

Create contractionary scenario data frame

```

contraction_df <- tibble(
  M2_growth = seq(-5, 5, by = 1),
  M2_growth_lag3 = seq(-5, 5, by = 1), # assume lag3 follows same trend
  Inflation_lag1 = last_infl_lag1      # constant lagged inflation
)

```

Generate predicted inflation and 95% confidence intervals

```

pred_con <- predict(model_multi, newdata = contraction_df,
  interval = "confidence", level = 0.95)

```

Combine predictions with scenario data

```

contraction_df <- bind_cols(contraction_df, as_tibble(pred_con))

```



```

# View predicted values
print(contraction_df)

# Plot predicted inflation vs M2 growth with 95% confidence interval
contraction_df %>%
  ggplot(aes(x = M2_growth, y = fit)) +
  geom_line(color = "red", size = 1) +
  geom_ribbon(aes(ymin = lwr, ymax = upr), fill = "pink", alpha = 0.2) +
  labs(
    title = "Predicted Inflation under Contractionary M2 Growth (-5% to +5%)",
    x = "M2 Growth (%)",
    y = "Predicted Inflation (%)"
  ) +
  theme_minimal()

# 12. Comparative checks (Q11)
# Compare coefficients
coef_summary <- summary(model_multi)$coefficients
print(coef_summary)

# Compare predictions visually
library(ggplot2)

# Add a scenario label
expansion_df$Scenario <- "Expansionary"
contraction_df$Scenario <- "Contractionary"

# Combine

```

```
combined_df <- bind_rows(expansion_df, contraction_df)

# Plot predicted inflation under both scenarios
ggplot(combined_df, aes(x = M2_growth, y = fit, color = Scenario)) +
  geom_line(size = 1) +
  geom_ribbon(aes(ymin = lwr, ymax = upr, fill = Scenario), alpha = 0.2) +
  labs(title = "Predicted Inflation: Expansionary vs Contractionary M2 Growth",
        x = "M2 Growth (%)", y = "Predicted Inflation (%)") +
  theme_minimal()
```

NONLINEAR MODELLING: Polynomial Regression

Q(b) Fit a polynomial (quadratic) regression model

R code to estimate the polynomial model:

```
model_poly <- lm(
  Inflation ~ M2_growth + I(M2_growth^2) + Inflation_lag1,
  data = data
)
```

```
cat("\nQ(b) Polynomial Regression Model Summary:\n")
```

```
summary(model_poly)
```

Q(c) Model Comparison and Interpretation

Compare Adjusted R², AIC, and BIC

Extract comparison metrics

```
comparison_metrics <- data.frame(
```

```

Model = c("Multiple Linear", "Polynomial"),
Adj_R2 = c(summary(model_multi)$adj.r.squared, summary(model_poly)$adj.r.squared),
AIC = c(AIC(model_multi), AIC(model_poly)),
BIC = c(BIC(model_multi), BIC(model_poly))
)

print(comparison_metrics)

#Generate the Correct Visualisation

# 1. Create a grid of M2_Growth values for smooth plotting
M2_range <- seq(min(data$M2_growth), max(data$M2_growth), length.out = 100)

# 2. Create a new data frame for prediction
# CRITICAL: Hold the lagged term (Inflation_lag1) constant at its mean
mean_Inflation_lag1 <- mean(data$Inflation_lag1, na.rm = TRUE)
prediction_data_poly <- data.frame(
  M2_growth = M2_range,
  Inflation_lag1 = mean_Inflation_lag1
)

# 3. Get predicted values from the polynomial model
prediction_data_poly$Predicted_Inflation <- predict(model_poly, newdata = prediction_data_poly)

# 4. Plotting the actual data and the fitted polynomial curve
polynomial_plot <- ggplot(data, aes(x = M2_growth, y = Inflation)) +
  # Scatter plot of actual data points
  geom_point(color = "gray", alpha = 0.6) +
  # The smooth fitted polynomial curve

```

```
geom_line(data = prediction_data_poly, aes(y = Predicted_Inflation), color = "darkred", linewidth =  
1.5) +  
  labs(  
    title = "Q(b) Polynomial Regression: Predicted Inflation vs M2 Growth (Lagged Inflation at Mean)",  
    x = "M2 Growth (%)",  
    y = "Inflation (%)"  
  ) +  
  theme_minimal() +  
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))  
  
print(polynomial_plot)
```